Customer Data Validation Report

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TOTAL RECORDS	CLEAN
500	372
REJECTED	**WARNINGS**
128	79
CLEANING ACTIONS	
341	

Showing 236 of 577 total issues | ERROR: checked, WARNING: checked, CLEANED: unchecked

ROW	SEVERITY	RULE	FIELD	MESSAGE	ORIGINAL	CORRECTED
3	ERROR	R05_FUTURE_DOB	date_of_birth	Date of birth is in the future: 2045-06-15	2045-06-15	
4	WARNING	R26_AOV_OUTLIER	total_spend,num_orders	Average order value outlier: $1175.27 (spend=3525.81, orders=3)	aov=1175.27	
5	WARNING	R17_MISSING_LAST_NAME	last_name	Missing last name		
7	ERROR	R02_INVALID_EMAIL	email	Invalid email format: double @@ in 'michelle.young@@outlook.com'	michelle.young@@outlook.com	
8	WARNING	R26_AOV_OUTLIER	total_spend,num_orders	Average order value outlier: $4855.68 (spend=4855.68, orders=1)	aov=4855.68	
9	WARNING	R25_TIER_SPEND_MISMATCH	loyalty_tier,total_spend	Bronze tier with $4353.89 total spend (high for tier)	tier=Bronze, spend=4353.89	
10	ERROR	R05_FUTURE_DOB	date_of_birth	Date of birth is in the future: 2045-06-15	2045-06-15	
13	ERROR	R08_NEGATIVE_SPEND	total_spend	Negative total spend (sentinel value): -999.99	-999.99	
16	ERROR	R03_DUPLICATE_EMAIL	email	Duplicate email address: 'james_harris55@hotmail.com'	james_harris55@hotmail.com	
26	ERROR	R01_MISSING_EMAIL	email	Missing email address (required field)		
27	WARNING	R26_AOV_OUTLIER	total_spend,num_orders	Average order value outlier: $2744.21 (spend=2744.21, orders=1)	aov=2744.21	
30	ERROR	R01_MISSING_EMAIL	email	Missing email address (required field)		
30	ERROR	R04_MISSING_BOTH_NAMES	first_name,last_name	Missing both first name and last name		
30	ERROR	R10_MISSING_SIGNUP_DATE	signup_date	Missing signup date (required for customer timeline)		
35	WARNING	R25_TIER_SPEND_MISMATCH	loyalty_tier,total_spend	Bronze tier with $4999.54 total spend (high for tier)	tier=Bronze, spend=4999.54	
36	ERROR	R01_MISSING_EMAIL	email	Missing email address (required field)		
36	ERROR	R11_ORDER_BEFORE_SIGNUP	last_order_date	Last order date (2023-01-13) is before signup date (2023-04-26)	2023-01-13	
37	ERROR	R11_ORDER_BEFORE_SIGNUP	last_order_date	Last order date (2022-09-06) is before signup date (2023-04-13)	2022-09-06	

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38	WARNING	R25_TIER_SPEND_MISMATCH	loyalty_tier,total_spend	Bronze tier with $4249.38 total spend (high for tier)	tier=Bronze, spend=4249.38
39	ERROR	R02_INVALID_EMAIL	email	Invalid email format: 'not_an_email'	not_an_email
39	ERROR	R11_ORDER_BEFORE_SIGNUP	last_order_date	Last order date (2022-09-07) is before signup date (2022-10-11)	2022-09-07
41	ERROR	R01_MISSING_EMAIL	email	Missing email address (required field)	
44	ERROR	R11_ORDER_BEFORE_SIGNUP	last_order_date	Last order date (2022-04-27) is before signup date (2023-06-01)	2022-04-27
47	ERROR	R11_ORDER_BEFORE_SIGNUP	last_order_date	Last order date (2022-08-19) is before signup date (2023-03-08)	2022-08-19
48	ERROR	R05_FUTURE_DOB	date_of_birth	Date of birth is in the future: 2045-06-15	2045-06-15
48	ERROR	R11_ORDER_BEFORE_SIGNUP	last_order_date	Last order date (2022-02-10) is before signup date (2023-04-15)	2022-02-10
51	ERROR	R01_MISSING_EMAIL	email	Missing email address (required field)	
51	ERROR	R04_MISSING_BOTH_NAMES	first_name,last_name	Missing both first name and last name	
51	ERROR	R10_MISSING_SIGNUP_DATE	signup_date	Missing signup date (required for customer timeline)	
55	ERROR	R01_MISSING_EMAIL	email	Missing email address (required field)	
56	ERROR	R09_INVALID_ZIP	zip_code	Invalid zip code (not 5 digits after cleaning): '999999'	999999
59	WARNING	R25_TIER_SPEND_MISMATCH	loyalty_tier,total_spend	Bronze tier with $4495.55 total spend (high for tier)	tier=Bronze, spend=4495.55
63	WARNING	R25_TIER_SPEND_MISMATCH	loyalty_tier,total_spend	Platinum tier with only $79.45 total spend	tier=Platinum, spend=79.45
64	ERROR	R01_MISSING_EMAIL	email	Missing email address (required field)	
65	ERROR	R01_MISSING_EMAIL	email	Missing email address (required field)	
67	ERROR	R11_ORDER_BEFORE_SIGNUP	last_order_date	Last order date (2022-05-09) is before signup date (2023-06-12)	2022-05-09
70	WARNING	R27_PHONE_MISSING_AREA_CODE	phone	Phone number appears to be missing area code (7 digits): '555-1234'	555-1234
73	ERROR	R02_INVALID_EMAIL	email	Invalid email format: 'not_an_email'	not_an_email
73	ERROR	R11_ORDER_BEFORE_SIGNUP	last_order_date	Last order date (2023-06-04) is before signup date (2023-06-07)	2023-06-04
79	WARNING	R25_TIER_SPEND_MISMATCH	loyalty_tier,total_spend	Bronze tier with $4831.34 total spend (high for tier)	tier=Bronze, spend=4831.34
80	ERROR	R02_INVALID_EMAIL	email	Invalid email format: 'not_an_email'	not_an_email
81	WARNING	R27_PHONE_MISSING_AREA_CODE	phone	Phone number appears to be missing area code (7 digits): '555-1234'	555-1234
82	ERROR	R01_MISSING_EMAIL	email	Missing email address (required field)	
86	ERROR	R02_INVALID_EMAIL	email	Invalid email format: double @@ in 'kennethrivera49@@hotmail.com'	kennethrivera49@@hotmail.com

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87	ERROR	R03_DUPLICATE_EMAIL	email	Duplicate email address: 'susan.perez@yahoo.com'	susan.perez@yahoo.com
88	ERROR	R02_INVALID_EMAIL	email	Invalid email format: 'matthewmartinez@livecom'	matthewmartinez@livecom
88	ERROR	R11_ORDER_BEFORE_SIGNUP	last_order_date	Last order date (2022-08-28) is before signup date (2023-06-09)	2022-08-28
89	ERROR	R02_INVALID_EMAIL	email	Invalid email format: 'donald_gonzalez60comcast.net'	donald_gonzalez60comcast.net
90	WARNING	R26_AOV_OUTLIER	total_spend,num_orders	Average order value outlier: $2212.15 (spend=4424.3, orders=2)	aov=2212.15
91	WARNING	R18_MISSING_PHONE	phone	Missing phone number	
92	ERROR	R02_INVALID_EMAIL	email	Invalid email format: double @@ in 'donald.lewis3@@outlook.com'	donald.lewis3@@outlook.com
95	WARNING	R20_MISSING_LOYALTY_TIER	loyalty_tier	Missing loyalty tier	
102	WARNING	R26_AOV_OUTLIER	total_spend,num_orders	Average order value outlier: $1976.98 (spend=3953.96, orders=2)	aov=1976.98
107	WARNING	R17_MISSING_LAST_NAME	last_name	Missing last name	
107	WARNING	R18_MISSING_PHONE	phone	Missing phone number	
108	ERROR	R02_INVALID_EMAIL	email	Invalid email format: 'williamcarter14outlook.com'	williamcarter14outlook.com
109	WARNING	R25_TIER_SPEND_MISMATCH	loyalty_tier,total_spend	Bronze tier with $4726.71 total spend (high for tier)	tier=Bronze, spend=4726.71
112	ERROR	R02_INVALID_EMAIL	email	Invalid email format: 'kimberlywhite66gmail.com'	kimberlywhite66gmail.com
112	ERROR	R11_ORDER_BEFORE_SIGNUP	last_order_date	Last order date (2022-05-24) is before signup date (2022-07-05)	2022-05-24
113	ERROR	R01_MISSING_EMAIL	email	Missing email address (required field)	
114	ERROR	R07_OVER_120	date_of_birth	Customer is over 120 (age 127, likely data error): 1899-01-01	1899-01-01
118	ERROR	R11_ORDER_BEFORE_SIGNUP	last_order_date	Last order date (2022-07-26) is before signup date (2023-03-08)	2022-07-26
120	WARNING	R25_TIER_SPEND_MISMATCH	loyalty_tier,total_spend	Bronze tier with $4540.61 total spend (high for tier)	tier=Bronze, spend=4540.61
121	WARNING	R19_MISSING_DOB	date_of_birth	Missing date of birth	
121	WARNING	R25_TIER_SPEND_MISMATCH	loyalty_tier,total_spend	Bronze tier with $4642.99 total spend (high for tier)	tier=Bronze, spend=4642.99
124	WARNING	R21_MISSING_PREFERRED_CONTACT	preferred_contact	Missing preferred contact method	
126	ERROR	R11_ORDER_BEFORE_SIGNUP	last_order_date	Last order date (2022-06-18) is before signup date (2023-01-24)	2022-06-18
128	ERROR	R11_ORDER_BEFORE_SIGNUP	last_order_date	Last order date (2022-07-04) is before signup date (2023-03-30)	2022-07-04
134	WARNING	R25_TIER_SPEND_MISMATCH	loyalty_tier,total_spend	Bronze tier with $4611.91 total spend (high for tier)	tier=Bronze, spend=4611.91
135	ERROR	R11_ORDER_BEFORE_SIGNUP	last_order_date	Last order date (2022-05-01) is before signup date (2023-05-14)	2022-05-01

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141	ERROR	R11_ORDER_BEFORE_SIGNUP	last_order_date	Last order date (2022-08-20) is before signup date (2023-05-19)	2022-08-20
142	ERROR	R01_MISSING_EMAIL	email	Missing email address (required field)	
143	WARNING	R17_MISSING_LAST_NAME	last_name	Missing last name	
144	ERROR	R11_ORDER_BEFORE_SIGNUP	last_order_date	Last order date (2022-02-24) is before signup date (2022-03-28)	2022-02-24
146	ERROR	R15_POSITIVE_ORDERS_NO_SPEND	num_orders,total_spend	Positive orders (21) but zero or missing spend	num_orders=21, total_spend=0
149	WARNING	R20_MISSING_LOYALTY_TIER	loyalty_tier	Missing loyalty tier	
151	ERROR	R01_MISSING_EMAIL	email	Missing email address (required field)	
152	WARNING	R19_MISSING_DOB	date_of_birth	Missing date of birth	
155	ERROR	R15_POSITIVE_ORDERS_NO_SPEND	num_orders,total_spend	Positive orders (29) but zero or missing spend	num_orders=29, total_spend=0
160	WARNING	R26_AOV_OUTLIER	total_spend,num_orders	Average order value outlier: $4328.96 (spend=4328.96, orders=1)	aov=4328.96
163	ERROR	R09_INVALID_ZIP	zip_code	Invalid zip code (not 5 digits after cleaning): 'ABCDE'	ABCDE
167	ERROR	R02_INVALID_EMAIL	email	Invalid email format: 'not_an_email'	not_an_email
169	ERROR	R02_INVALID_EMAIL	email	Invalid email format: 'not_an_email'	not_an_email
172	ERROR	R11_ORDER_BEFORE_SIGNUP	last_order_date	Last order date (2022-10-11) is before signup date (2023-05-27)	2022-10-11
174	ERROR	R05_FUTURE_DOB	date_of_birth	Date of birth is in the future: 2045-06-15	2045-06-15
174	ERROR	R08_NEGATIVE_SPEND	total_spend	Negative total spend (sentinel value): -999.99	-999.99
179	WARNING	R20_MISSING_LOYALTY_TIER	loyalty_tier	Missing loyalty tier	
180	ERROR	R08_NEGATIVE_SPEND	total_spend	Negative total spend (sentinel value): -999.99	-999.99
184	ERROR	R01_MISSING_EMAIL	email	Missing email address (required field)	
184	ERROR	R04_MISSING_BOTH_NAMES	first_name,last_name	Missing both first name and last name	
184	ERROR	R10_MISSING_SIGNUP_DATE	signup_date	Missing signup date (required for customer timeline)	
188	ERROR	R11_ORDER_BEFORE_SIGNUP	last_order_date	Last order date (2023-03-20) is before signup date (2023-04-14)	2023-03-20
190	ERROR	R01_MISSING_EMAIL	email	Missing email address (required field)	
195	WARNING	R19_MISSING_DOB	date_of_birth	Missing date of birth	
199	ERROR	R11_ORDER_BEFORE_SIGNUP	last_order_date	Last order date (2022-07-30) is before signup date (2022-12-28)	2022-07-30
201	ERROR	R02_INVALID_EMAIL	email	Invalid email format: double @@ in 'kevin.green@@outlook.com'	kevin.green@@outlook.com
202	ERROR	R08_NEGATIVE_SPEND	total_spend	Negative total spend (sentinel value): -50.0	-50.0

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203	WARNING	R20_MISSING_LOYALTY_TIER	loyalty_tier	Missing loyalty tier	
204	WARNING	R25_TIER_SPEND_MISMATCH	loyalty_tier,total_spend	Bronze tier with $4941.52 total spend (high for tier)	tier=Bronze, spend=4941.52
205	ERROR	R02_INVALID_EMAIL	email	Invalid email format: 'thomas_lee72aol.com'	thomas_lee72aol.com
207	WARNING	R25_TIER_SPEND_MISMATCH	loyalty_tier,total_spend	Bronze tier with $4862.76 total spend (high for tier)	tier=Bronze, spend=4862.76
209	WARNING	R25_TIER_SPEND_MISMATCH	loyalty_tier,total_spend	Bronze tier with $4590.98 total spend (high for tier)	tier=Bronze, spend=4590.98
211	ERROR	R02_INVALID_EMAIL	email	Invalid email format: 'not_an_email'	not_an_email
212	ERROR	R01_MISSING_EMAIL	email	Missing email address (required field)	
214	WARNING	R17_MISSING_LAST_NAME	last_name	Missing last name	
218	ERROR	R11_ORDER_BEFORE_SIGNUP	last_order_date	Last order date (2022-04-01) is before signup date (2022-04-15)	2022-04-01
219	ERROR	R11_ORDER_BEFORE_SIGNUP	last_order_date	Last order date (2022-07-15) is before signup date (2023-04-30)	2022-07-15
220	WARNING	R25_TIER_SPEND_MISMATCH	loyalty_tier,total_spend	Bronze tier with $4120.98 total spend (high for tier)	tier=Bronze, spend=4120.98
221	ERROR	R11_ORDER_BEFORE_SIGNUP	last_order_date	Last order date (2022-05-15) is before signup date (2023-04-26)	2022-05-15
228	ERROR	R02_INVALID_EMAIL	email	Invalid email format: 'brian.walkerhotmail.com'	brian.walkerhotmail.com
229	ERROR	R02_INVALID_EMAIL	email	Invalid email format: 'george.rodriguez53gmail.com'	george.rodriguez53gmail.com
230	WARNING	R16_MISSING_FIRST_NAME	first_name	Missing first name	
234	ERROR	R11_ORDER_BEFORE_SIGNUP	last_order_date	Last order date (2022-02-17) is before signup date (2022-08-16)	2022-02-17
235	ERROR	R09_INVALID_ZIP	zip_code	Invalid zip code (not 5 digits after cleaning): 'ABCDE'	ABCDE
235	ERROR	R11_ORDER_BEFORE_SIGNUP	last_order_date	Last order date (2022-04-17) is before signup date (2022-05-19)	2022-04-17
238	ERROR	R02_INVALID_EMAIL	email	Invalid email format: 'andrew_nelsonyahoo.com'	andrew_nelsonyahoo.com
242	ERROR	R11_ORDER_BEFORE_SIGNUP	last_order_date	Last order date (2022-03-11) is before signup date (2022-05-25)	2022-03-11
244	ERROR	R05_FUTURE_DOB	date_of_birth	Date of birth is in the future: 2045-06-15	2045-06-15
246	WARNING	R18_MISSING_PHONE	phone	Missing phone number	
250	ERROR	R02_INVALID_EMAIL	email	Invalid email format: 'not_an_email'	not_an_email
251	WARNING	R25_TIER_SPEND_MISMATCH	loyalty_tier,total_spend	Bronze tier with $4501.59 total spend (high for tier)	tier=Bronze, spend=4501.59
253	ERROR	R11_ORDER_BEFORE_SIGNUP	last_order_date	Last order date (2022-03-20) is before signup date (2022-10-25)	2022-03-20
254	ERROR	R01_MISSING_EMAIL	email	Missing email address (required field)	
256	WARNING	R19_MISSING_DOB	date_of_birth	Missing date of birth	

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259	WARNING	R20_MISSING_LOYALTY_TIER	loyalty_tier	Missing loyalty tier	
260	ERROR	R01_MISSING_EMAIL	email	Missing email address (required field)	
264	ERROR	R03_DUPLICATE_EMAIL	email	Duplicate email address: 'barbaraking52@hotmail.com'	barbaraking52@hotmail.com
265	WARNING	R19_MISSING_DOB	date_of_birth	Missing date of birth	
268	ERROR	R07_OVER_120	date_of_birth	Customer is over 120 (age 127, likely data error): 1899-01-01	1899-01-01
268	ERROR	R11_ORDER_BEFORE_SIGNUP	last_order_date	Last order date (2022-02-28) is before signup date (2023-03-22)	2022-02-28
272	ERROR	R08_NEGATIVE_SPEND	total_spend	Negative total spend (sentinel value): -1.99	-1.99
274	ERROR	R11_ORDER_BEFORE_SIGNUP	last_order_date	Last order date (2022-05-03) is before signup date (2022-09-06)	2022-05-03
277	ERROR	R01_MISSING_EMAIL	email	Missing email address (required field)	
279	ERROR	R11_ORDER_BEFORE_SIGNUP	last_order_date	Last order date (2022-09-14) is before signup date (2023-01-04)	2022-09-14
285	ERROR	R01_MISSING_EMAIL	email	Missing email address (required field)	
287	WARNING	R26_AOV_OUTLIER	total_spend,num_orders	Average order value outlier: $2989.05 (spend=2989.05, orders=1)	aov=2989.05
293	WARNING	R16_MISSING_FIRST_NAME	first_name	Missing first name	
296	WARNING	R16_MISSING_FIRST_NAME	first_name	Missing first name	
301	WARNING	R18_MISSING_PHONE	phone	Missing phone number	
302	WARNING	R17_MISSING_LAST_NAME	last_name	Missing last name	
304	ERROR	R09_INVALID_ZIP	zip_code	Invalid zip code (not 5 digits after cleaning): 'ABCDE'	ABCDE
308	ERROR	R02_INVALID_EMAIL	email	Invalid email format: 'donald_torrescomcast.net'	donald_torrescomcast.net
311	WARNING	R19_MISSING_DOB	date_of_birth	Missing date of birth	
311	WARNING	R21_MISSING_PREFERRED_CONTACT	preferred_contact	Missing preferred contact method	
312	ERROR	R02_INVALID_EMAIL	email	Invalid email format: 'donald_flores@icloudcom'	donald_flores@icloudcom
312	ERROR	R11_ORDER_BEFORE_SIGNUP	last_order_date	Last order date (2022-01-09) is before signup date (2022-06-05)	2022-01-09
318	WARNING	R19_MISSING_DOB	date_of_birth	Missing date of birth	
321	ERROR	R03_DUPLICATE_EMAIL	email	Duplicate email address: 'steven.mitchell78@gmail.com'	steven.mitchell78@gmail.com
322	ERROR	R03_DUPLICATE_EMAIL	email	Duplicate email address: 'richard_jones15@live.com'	richard_jones15@live.com
323	ERROR	R01_MISSING_EMAIL	email	Missing email address (required field)	
324	WARNING	R25_TIER_SPEND_MISMATCH	loyalty_tier,total_spend	Bronze tier with $4493.72 total spend (high for tier)	tier=Bronze, spend=4493.72

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325	WARNING	R26_AOV_OUTLIER	total_spend,num_orders	Average order value outlier: $3010.07 (spend=3010.07, orders=1)	aov=3010.07
326	ERROR	R11_ORDER_BEFORE_SIGNUP	last_order_date	Last order date (2022-02-20) is before signup date (2022-09-13)	2022-02-20
331	WARNING	R25_TIER_SPEND_MISMATCH	loyalty_tier,total_spend	Bronze tier with $4161.89 total spend (high for tier)	tier=Bronze, spend=4161.89
332	WARNING	R18_MISSING_PHONE	phone	Missing phone number	
332	WARNING	R26_AOV_OUTLIER	total_spend,num_orders	Average order value outlier: $1390.16 (spend=2780.33, orders=2)	aov=1390.16
340	WARNING	R19_MISSING_DOB	date_of_birth	Missing date of birth	
341	WARNING	R19_MISSING_DOB	date_of_birth	Missing date of birth	
342	WARNING	R19_MISSING_DOB	date_of_birth	Missing date of birth	
343	ERROR	R01_MISSING_EMAIL	email	Missing email address (required field)	
344	ERROR	R03_DUPLICATE_EMAIL	email	Duplicate email address: 'susan.rodriguez@live.com'	susan.rodriguez@live.com
345	ERROR	R02_INVALID_EMAIL	email	Invalid email format: 'amanda.thompson79hotmail.com'	amanda.thompson79hotmail.com
345	ERROR	R11_ORDER_BEFORE_SIGNUP	last_order_date	Last order date (2022-04-23) is before signup date (2023-06-06)	2022-04-23
348	ERROR	R02_INVALID_EMAIL	email	Invalid email format: double @@ in 'elizabethrobinson58@@aol.com'	elizabethrobinson58@@aol.com
349	ERROR	R08_NEGATIVE_SPEND	total_spend	Negative total spend (sentinel value): -999.99	-999.99
350	ERROR	R03_DUPLICATE_EMAIL	email	Duplicate email address: 'mark.carter68@comcast.net'	mark.carter68@comcast.net
351	ERROR	R05_FUTURE_DOB	date_of_birth	Date of birth is in the future: 2045-06-15	2045-06-15
356	WARNING	R18_MISSING_PHONE	phone	Missing phone number	
358	WARNING	R27_PHONE_MISSING_AREA_CODE	phone	Phone number appears to be missing area code (7 digits): '555-1234'	555-1234
359	ERROR	R01_MISSING_EMAIL	email	Missing email address (required field)	
360	ERROR	R08_NEGATIVE_SPEND	total_spend	Negative total spend (sentinel value): -999.99	-999.99
361	ERROR	R09_INVALID_ZIP	zip_code	Invalid zip code (not 5 digits after cleaning): '1234'	1234
364	WARNING	R26_AOV_OUTLIER	total_spend,num_orders	Average order value outlier: $1313.67 (spend=3941.01, orders=3)	aov=1313.67
367	ERROR	R02_INVALID_EMAIL	email	Invalid email format: 'not_an_email'	not_an_email
371	ERROR	R01_MISSING_EMAIL	email	Missing email address (required field)	
372	ERROR	R03_DUPLICATE_EMAIL	email	Duplicate email address: 'susan.perez@yahoo.com'	susan.perez@yahoo.com
373	ERROR	R01_MISSING_EMAIL	email	Missing email address (required field)	
374	ERROR	R08_NEGATIVE_SPEND	total_spend	Negative total spend (sentinel value): -999.99	-999.99

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379	ERROR	R11_ORDER_BEFORE_SIGNUP	last_order_date	Last order date (2023-01-28) is before signup date (2023-04-21)	2023-01-28
383	ERROR	R01_MISSING_EMAIL	email	Missing email address (required field)	
385	WARNING	R25_TIER_SPEND_MISMATCH	loyalty_tier,total_spend	Bronze tier with $4968.93 total spend (high for tier)	tier=Bronze, spend=4968.93
386	ERROR	R11_ORDER_BEFORE_SIGNUP	last_order_date	Last order date (2023-02-19) is before signup date (2023-06-21)	2023-02-19
387	WARNING	R18_MISSING_PHONE	phone	Missing phone number	
388	ERROR	R01_MISSING_EMAIL	email	Missing email address (required field)	
389	WARNING	R25_TIER_SPEND_MISMATCH	loyalty_tier,total_spend	Bronze tier with $4543.15 total spend (high for tier)	tier=Bronze, spend=4543.15
392	ERROR	R01_MISSING_EMAIL	email	Missing email address (required field)	
394	ERROR	R07_OVER_120	date_of_birth	Customer is over 120 (age 127, likely data error): 1899-01-01	1899-01-01
396	WARNING	R26_AOV_OUTLIER	total_spend,num_orders	Average order value outlier: $4118.34 (spend=4118.34, orders=1)	aov=4118.34
396	WARNING	R27_PHONE_MISSING_AREA_CODE	phone	Phone number appears to be missing area code (7 digits): '555-1234'	555-1234
401	ERROR	R11_ORDER_BEFORE_SIGNUP	last_order_date	Last order date (2023-05-02) is before signup date (2023-06-17)	2023-05-02
412	WARNING	R25_TIER_SPEND_MISMATCH	loyalty_tier,total_spend	Bronze tier with $4928.97 total spend (high for tier)	tier=Bronze, spend=4928.97
419	ERROR	R03_DUPLICATE_EMAIL	email	Duplicate email address: 'carolwright92@gmail.com'	carolwright92@gmail.com
420	WARNING	R20_MISSING_LOYALTY_TIER	loyalty_tier	Missing loyalty tier	
423	WARNING	R21_MISSING_PREFERRED_CONTACT	preferred_contact	Missing preferred contact method	
427	ERROR	R03_DUPLICATE_EMAIL	email	Duplicate email address: 'mary.scott@icloud.com'	mary.scott@icloud.com
428	WARNING	R19_MISSING_DOB	date_of_birth	Missing date of birth	
429	ERROR	R03_DUPLICATE_EMAIL	email	Duplicate email address: 'thomasyoung87@icloud.com'	thomasyoung87@icloud.com
430	ERROR	R02_INVALID_EMAIL	email	Invalid email format: 'donaldrodriguez28@comcastnet'	donaldrodriguez28@comcastnet
430	ERROR	R09_INVALID_ZIP	zip_code	Invalid zip code (not 5 digits after cleaning): 'ABCDE'	ABCDE
431	ERROR	R11_ORDER_BEFORE_SIGNUP	last_order_date	Last order date (2022-06-23) is before signup date (2023-06-06)	2022-06-23
433	WARNING	R20_MISSING_LOYALTY_TIER	loyalty_tier	Missing loyalty tier	
435	ERROR	R11_ORDER_BEFORE_SIGNUP	last_order_date	Last order date (2022-04-07) is before signup date (2023-03-01)	2022-04-07
437	WARNING	R17_MISSING_LAST_NAME	last_name	Missing last name	
438	ERROR	R09_INVALID_ZIP	zip_code	Invalid zip code (not 5 digits after cleaning): '999999'	999999

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444	WARNING	R25_TIER_SPEND_MISMATCH	loyalty_tier,total_spend	Bronze tier with $4081.68 total spend (high for tier)	tier=Bronze, spend=4081.68
446	ERROR	R07_OVER_120	date_of_birth	Customer is over 120 (age 127, likely data error): 1899-01-01	1899-01-01
447	WARNING	R17_MISSING_LAST_NAME	last_name	Missing last name	
448	WARNING	R19_MISSING_DOB	date_of_birth	Missing date of birth	
449	ERROR	R11_ORDER_BEFORE_SIGNUP	last_order_date	Last order date (2022-03-30) is before signup date (2022-06-26)	2022-03-30
453	ERROR	R01_MISSING_EMAIL	email	Missing email address (required field)	
456	WARNING	R25_TIER_SPEND_MISMATCH	loyalty_tier,total_spend	Bronze tier with $4136.75 total spend (high for tier)	tier=Bronze, spend=4136.75
463	ERROR	R01_MISSING_EMAIL	email	Missing email address (required field)	
464	WARNING	R19_MISSING_DOB	date_of_birth	Missing date of birth	
466	WARNING	R25_TIER_SPEND_MISMATCH	loyalty_tier,total_spend	Bronze tier with $4786.89 total spend (high for tier)	tier=Bronze, spend=4786.89
467	ERROR	R01_MISSING_EMAIL	email	Missing email address (required field)	
468	WARNING	R20_MISSING_LOYALTY_TIER	loyalty_tier	Missing loyalty tier	
468	WARNING	R27_PHONE_MISSING_AREA_CODE	phone	Phone number appears to be missing area code (7 digits): '555-1234'	555-1234
471	ERROR	R01_MISSING_EMAIL	email	Missing email address (required field)	
472	ERROR	R11_ORDER_BEFORE_SIGNUP	last_order_date	Last order date (2023-01-25) is before signup date (2023-03-27)	2023-01-25
474	ERROR	R11_ORDER_BEFORE_SIGNUP	last_order_date	Last order date (2022-01-10) is before signup date (2023-02-15)	2022-01-10
476	ERROR	R01_MISSING_EMAIL	email	Missing email address (required field)	
476	ERROR	R04_MISSING_BOTH_NAMES	first_name,last_name	Missing both first name and last name	
476	ERROR	R10_MISSING_SIGNUP_DATE	signup_date	Missing signup date (required for customer timeline)	
477	ERROR	R05_FUTURE_DOB	date_of_birth	Date of birth is in the future: 2045-06-15	2045-06-15
478	WARNING	R25_TIER_SPEND_MISMATCH	loyalty_tier,total_spend	Platinum tier with only $19.76 total spend	tier=Platinum, spend=19.76
478	WARNING	R26_AOV_OUTLIER	total_spend,num_orders	Average order value outlier: $0.71 (spend=19.76, orders=28)	aov=0.71
479	WARNING	R19_MISSING_DOB	date_of_birth	Missing date of birth	
480	ERROR	R02_INVALID_EMAIL	email	Invalid email format: 'williamcarter14outlook.com'	williamcarter14 outlook.com
483	ERROR	R05_FUTURE_DOB	date_of_birth	Date of birth is in the future: 2045-06-15	2045-06-15
486	WARNING	R16_MISSING_FIRST_NAME	first_name	Missing first name	
488	ERROR	R11_ORDER_BEFORE_SIGNUP	last_order_date	Last order date (2022-08-13) is before signup date (2023-06-21)	2022-08-13

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491	WARNING	R21_MISSING_PREFERRED_CONTACT	preferred_contact	Missing preferred contact method
493	WARNING	R18_MISSING_PHONE	phone	Missing phone number
500	ERROR	R01_MISSING_EMAIL	email	Missing email address (required field)
500	ERROR	R04_MISSING_BOTH_NAMES	first_name,last_name	Missing both first name and last name
500	ERROR	R10_MISSING_SIGNUP_DATE	signup_date	Missing signup date (required for customer timeline)